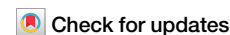


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AI-assisted multi-modal information for the screening of depression: a systematic review and meta-analysis



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Depression is a prevalent and costly mental disorder across all ages. Artificial intelligence (AI)-assisted physiological and behavioral information—such as electroencephalography (EEG), eye movement, video or audio monitoring, and gait analysis—offers a promising tool for depression screening. We systematically reviewed the classification performance of these AI-assisted measures in depression screening. A comprehensive literature search was conducted in Google Scholar, Web of Science, and IEEE Xplore, with the search date up to June 7, 2025. The reported AUC values are pooled estimates calculated from all results of eligible studies. AI-assisted multi-modal methods achieved a pooled AUC of 0.95 (95% CI: 0.92–0.96), outperforming uni-modal methods (pooled AUC: 0.84–0.92). Subgroup analysis indicated deep learning models showed higher performance, with an AUC of 0.95 (95% CI: 0.93–0.97). These findings highlight the potential of AI-based multi-modal information in depression screening and emphasize the need to establish standardized databases and improve research design.

Depression is a leading global mental health disorder and a major contributor to the worldwide burden of disability¹. The potential pathogenesis of depression remains unclear, and reliable screening in clinical practice is challenging, as current assessments rely primarily on patient history and self-reports². In addition, this makes large-scale community screening difficult and increases the risk of missed diagnoses. Therefore, effective and accurate methods for depression diagnosis are essential for early intervention. Recently, with advances in medical technology, cost-effective, non-invasive, and easy to collect techniques—such as EEG, eye movement, video or audio monitoring, and gait analysis—have been increasingly implemented in depression screening^{3–6}. These methods provide objective indicators for physiological and behavioral information.

Depression is a clinically and etiologically heterogeneous disorder⁷, and this heterogeneity is reflected in the variability of objectively measurable biological markers detected through physiological and behavioral assessments. Based on EEG facial recognition task or attention task, researchers detected abnormal brain activation in emotional networks in depression patients, including the dorsolateral prefrontal cortex (DLPFC), anterior cingulate cortex (ACC), orbitofrontal cortex (OFC), and amygdala^{8–11}. In addition, researches of eye movement tracking reveal that depression patients tend to focus more on negative stimuli while ignoring positive

interactions when viewing emotional images^{12–14}. This may be accompanied by changes in the frontal eye field (FEF), parietal cortex, and occipital cortex. Additionally, depression patients exhibit slow audio rates and increased negative facial expressions¹⁵. Furthermore, there is a bidirectional interaction between sensorimotor system and brain networks related to emotions and higher cognitive function^{16,17}. Depression may result in severely debilitating conditions accompanied with abnormal gait¹⁸. We summarized underlying brain areas related to different physiological and behavioral information in Fig. 1. However, due to the multifaceted nature of the symptoms of depression, a uni-modal physiological or behavioral information may not be sufficient to capture enough of them for effective screening¹⁹.

Current research shows that multi-modal information can lead to more personalized and accurate screening²⁰. However, the challenge of exploring the correlations and complementarity between multi-modal information is an urgent problem to be solved. With the development of artificial intelligence (AI), researchers are trying to use more automated, objective, efficient, and real-time methods to depression screening. The AI technology especially deep learning (DL), promises to enhance the accuracy of multi-modal feature integration, thereby enhancing classification performance and reducing screening errors. Although numerous studies have

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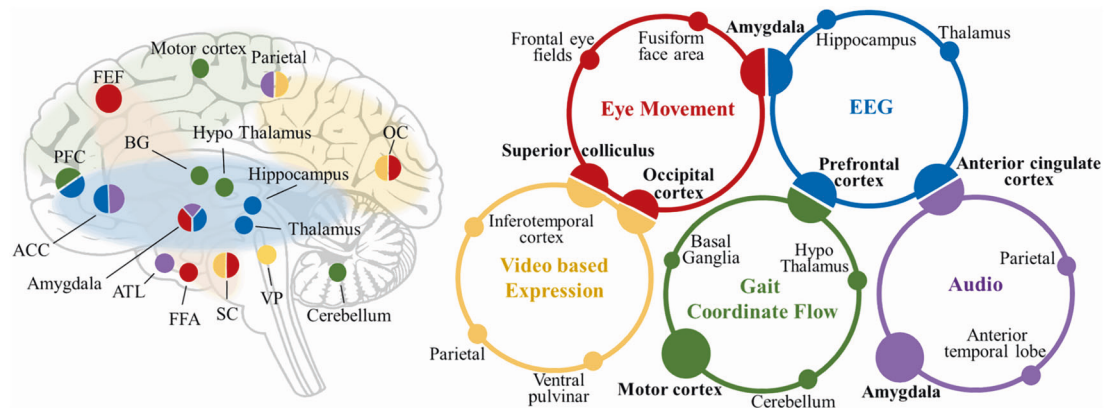


Fig. 1 | Brain activation patterns in depression across key functional domains. Brain activation patterns in depression across key functional domains, including altered emotional processing, negative attentional bias, delayed auditory responses,

and impaired sensorimotor-emotional integration. BG Basal Ganglia, ATL Anterior Temporal Lobe, FFA Fusiform Face Area, VP Ventral Pallidum, SC Superior Colliculus, OC Occipital Cortex.

employed various AI algorithms^{20–22}, comprehensive meta-analyses using multiple modalities for depression screening within digital healthcare is still insufficient.

This study systematically examines whether integrating AI-assisted (i.e., employing artificial intelligence algorithms, including machine learning and deep learning, for feature extraction and classification) multi-modal physiological and behavioral information can improve the classification performance for depression screening. Multi-modal AI-assisted screening methods have already achieved significant breakthroughs in other areas of medicine^{23,24}, which suggests promising potential for mental health applications as well. Although prior meta-analyses has explored digital signals from wearable devices (such as physical activity data, location data, etc.)²¹ and social media platforms (including tweets and communication logs)^{25,26} as potential indicators of depression. In contrast, our meta-analyses concentrate on objectively measurable physiological and behavioral information that are commonly used in fusion frameworks, such as EEG, eye movement, facial expressions, speech, and gait. By conducting a systematic meta-analysis of the modeling results derived from published data, we seek to highlight the unique value of multi-modal integration and outline future directions for more effective and objective depression screening.

Results

Study selection and characteristics of eligible studies

A total of 1564 records were identified in the initial search, of which 981 were duplicates. After screening the titles and abstracts, 108 studies were excluded as they were clearly unrelated to the research topic, leaving 475 articles for full-text eligibility assessment. Among these, 397 were excluded after reviewing the content of the full-text manuscripts, resulting in 80 studies included in the qualitative synthesis (comprising 33 EEG studies, 6 eye movement studies, 6 video studies, 16 audio studies, 6 gait studies, and 13 multi-modal studies). These 80 studies^{27–106} provided sufficient data to meet the inclusion criteria for the meta-analysis (Fig. 2). The detailed characteristics of these included studies are shown in Table 1 and Supplementary Tables 1–6. All studies used retrospective data.

71 of 80 studies (89%) explicitly reported that the ground truth assessment of their classification labels came from scales (e.g., DSM-IV and -V, PHQ-8, and -9, etc). The remaining nine studies did not specify any depression diagnosis or screening criteria. Furthermore, data were sourced from public datasets in 38 out of 80 studies (47.5%), while the remaining studies involved the collection of proprietary datasets (42/80, 52.5%). The predominant ground truth assessments utilized were the Diagnostic and Statistical Manual of Mental Disorders, fourth/fifth edition (25/80, 31.3%), and the Nine/Eight-item Patient Health Questionnaire (26/80, 32.5%). Tasks assigned to subjects under each modality (including both uni-modal

studies and multi-modal studies containing the corresponding modality) were eye open/closure for EEG (24/38, 63.1%), clinical interviews for video (8/16, 50.0%) and audio (23/30, 76.7%), free viewing for eye movement (7/7, 100.0%), and free walking for gait (4/6, 66.7%). The studies collectively employed 19 algorithms, with the most frequently utilized ones being support vector machine (SVM) (36/80, 45.0%), k-nearest neighbor (15/80, 18.8%), and convolutional neural network (CNN) (17/80, 21.3%).

In the multi-modal depression screening research, 13 studies provided a total of 15 investigations on different types of modal fusion methods and four results. Four of investigations used EEG and audio information (4/15, 26.7%), one used EEG and eye movement (1/15, 6.7%), and the remaining 8 investigations used audio, video, and text information (10/15, 66.7%).

Pooled performance of AI algorithms for classifying Depression from NC

Table 2 summarizes the comprehensive performance estimation of AI-assisted screening of various modal information for depression. Forest plots can be found in the Supplementary Figs. 1–12, and SROC curves can be found in Supplementary Figs. 13–18.

Thirty-three studies involving the screening of depression by EEG provided sufficient data to construct eighty-eight contingency tables and determine classification performance indicators. The pooled estimates for this group were 0.85 SE (95% CI: 0.84–0.87), 0.86 SP (95% CI: 0.84–0.88), and an AUC of 0.92 (95% CI: 0.89–0.94) (Fig. 3). When the contingency table with the highest performance was selected, yielded a pooled SE of 0.86 (95% CI: 0.83–0.89), SP of 0.89 (95% CI: 0.85–0.92), and AUC of 0.94 (95% CI: 0.92–0.96).

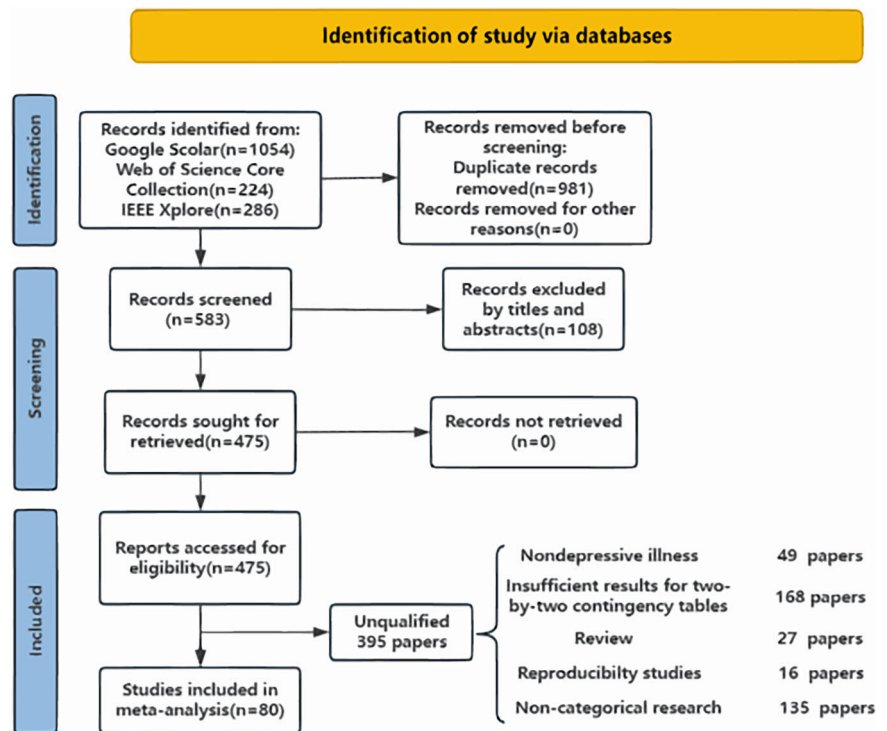
A total of 24 contingency tables from six studies on eye movement depression screening, all utilizing the ML algorithm. The pooled estimates for this group were 0.77 SE (95% CI: 0.72–0.81), 0.86 SP (95% CI: 0.83–0.89), and an AUC of 0.88 (95% CI: 0.85–0.91) (Fig. 3). When the contingency table with the highest performance was selected, yielded a pooled SE of 0.81 (95% CI: 0.74–0.87), SP of 0.85 (95% CI: 0.77–0.90), and AUC of 0.90 (95% CI: 0.87–0.92).

Six studies on depression screening of video provide 13 contingency tables. The pooled estimates for this group were 0.83 SE (95% CI: 0.78–0.87), 0.75 SP (95% CI: 0.70–0.80), and an AUC of 0.86 (95% CI: 0.82–0.88) (Fig. 3). When the contingency table with the highest performance was selected, yielded a pooled SE of 0.82 (95% CI: 0.72–0.89), SP of 0.78 (95% CI: 0.66–0.87), and AUC of 0.87 (95% CI: 0.84–0.90).

Sixteen studies on depression screening of audio provided a total of 28 contingency tables. The pooled estimates for this group were 0.86 SE (95% CI: 0.81–0.89), 0.86 SP (95% CI: 0.81–0.90), and an AUC of 0.92 (95% CI: 0.90–0.94) (Fig. 3). When the contingency table with the

Fig. 2 | PRISMA flowchart of study selection.

Illustration of the identification, screening, eligibility assessment, and inclusion of studies in the systematic review.



highest performance was selected, yielded a pooled SE of 0.86 (95% CI: 0.80–0.91), SP of 0.90 (95% CI: 0.86–0.93), and AUC of 0.95 (95% CI: 0.92–0.96).

A total of 25 contingency tables from six studies on gait depression screening. The pooled estimates for this group were 0.79 SE (95% CI: 0.73–0.84), 0.77 SP (95% CI: 0.73–0.81), and an AUC of 0.84 (95% CI: 0.81–0.87) (Fig. 3). When the contingency table with the highest performance was selected, yielded a pooled SE of 0.85 (95% CI: 0.75–0.91), SP of 0.83 (95% CI: 0.78–0.88), and AUC of 0.90 (95% CI: 0.87–0.92).

For all multi-modal depression screening studies, the data provided by 39 contingency tables in 13 studies created ideal results. The pooled estimates for this group were 0.88 SE (95% CI: 0.85–0.91), 0.90 SP (95% CI: 0.88–0.91), and an AUC of 0.95 (95% CI: 0.92–0.96) (Fig. 3). When the contingency table with the highest performance was selected, yielded a pooled SE of 0.89 (95% CI: 0.84–0.93), SP of 0.91 (95% CI: 0.88–0.94), and AUC of 0.96 (95% CI: 0.94–0.97).

Heterogeneity analysis

To explore the heterogeneity among studies, we performed further subgroup analyses of studies on each modality based on the features or signals in the input classifier (Table 3).

Low heterogeneity was observed in the depression classification of the eye movement study, with an I² of 33.15% (95% CI: 0.00–66.40) for SE and an I² of 47.67% (95% CI: 22.76–72.57) for SP. Using EEG technology, low to moderate heterogeneity was observed when distinguishing depression from NC, with an I² of 32.64% (95% CI: 14.76–50.51) for SE and an I² of 56.14% (95% CI: 45.67–66.61) for SP. Additionally, low to moderate heterogeneity was also observed in the depression classification of the multi-modal study, with an I² of 66.67% (95% CI: 55.53–77.82) for SE and an I² of 50.18% (95% CI: 31.74–68.61) for SP. However, moderate to high heterogeneity was observed across studies based on video, audio, and gait. In comparison, in 19 subgroup analyses across all 6 modalities, for a total of 38 Low to moderate heterogeneity was generally observed in the I² values of SE and SP. Specifically, there were 37 groups with I² values less than 75%.

The results revealed statistically significant differences in covariates. Publication bias of groups and subgroups resulting from visual inspection of funnel plots is shown in Supplementary Figs. 19–24.

Quality assessment

The quality of the included studies was determined by QUADAS-2. Detailed assessment results are shown in a diagram in Supplementary Fig. 25. Nearly half of the included studies ($n = 38$) demonstrated high or unclear risk of bias in patient selection, due to insufficient information provided to verify whether appropriate continuous samples were used to select eligible patients.

Discussion

In recent years, automatic depression screening methods have become increasingly diverse, especially combining with multi-modal physiological and behavioral information. However, most existing studies have focused on single-modality approaches. For example, depression screening using wearable devices and AI has achieved classification accuracy ranging from 70% to 89%²¹, which is similar to the performance of single-modality approaches in our meta-analysis. Nevertheless, there is a significant disparity in data formats across different modalities for depression screening, and various AI technologies still face challenges in integrating multi-modal inputs effectively. Therefore, we aim to explore whether AI-assisted multi-modal information could enhance the classification performance for depression screening. Given these challenges, we aimed to explore whether AI-assisted multi-modal information could further enhance the classification performance for depression screening. By adhering strictly to screen review guidelines, we ensured the integrity of our study. While our research demonstrates that uni-modal analysis using either voice or EEG alone achieves satisfactory diagnostic performance, the integration of multiple modalities yields superior classification accuracy. Notably, AI-assisted multiple modalities can classify patients with and without depression in up to 96% of cases (Table 2).

Due to decades of rapid development, EEG and audio signals have become a crucial method for screening and treating mental disorders¹⁰⁷. The availability of relatively sufficient data now allows meta-analyses to validate

Table 1 | Characteristics of all included studies (n = 80)

Category	Feature	Number of studies(%)	References
Extracted features (EEG)	Power spectral density	7 (18.4)	28, 30, 34, 35, 46, 48,86
	Lempel–Ziv complexity	6 (15.8)	27, 35, 37, 48, 50,86
	Band power	6 (15.8)	29, 30, 35, 48, 52,55
	Variance	6 (15.8)	30, 35, 45, 46, 50,86
	Hjorth complexity	4 (10.5)	30, 35, 46,86
	Hjorth mobility	4 (10.5)	30, 35, 46,86
	Hjorth activity	4 (10.5)	30, 34, 35,86
	Skewness	4 (10.5)	30, 45, 46,50
	Kurtosis	4 (10.5)	30, 45, 46,50
	Correlation dimension	4 (10.5)	35, 42, 82,86
	C0-complexity	4 (10.5)	35, 42, 82,86
	Shannon entropy	3 (7.9)	30, 37,50
	Approximate entropy	3 (7.9)	35, 50,86
	Permutation entropy	3 (7.9)	35, 37,86
	Sample entropy	3 (7.9)	48, 50,52
	Mean frequency	2 (5.3)	30,37
	Node2vec graph features	2 (5.3)	31,38
	Mean Square	2 (5.3)	35,50
	Kolmogorov entropy	2 (5.3)	35,86
	Lyapunov exponent	2 (5.3)	35,86
	Singular-value deposition entropy	2 (5.3)	35,86
	Max of the Spectrum	2 (5.3)	42,82
	Mean of the Spectrum	2 (5.3)	42,82
	Renyi entropy	2 (5.3)	42,82
	Relative power	2 (5.3)	46,50
	Vigilance regulation	1 (2.6)	29
	Singular value	1 (2.6)	30
	Permutation entropy	1 (2.6)	35
	Centroid frequency	1 (2.6)	37
	Center of the Spectrum	1 (2.6)	42
	Energy spectral density	1 (2.6)	45
	Graph entropy	1 (2.6)	49
	Rhythm index	1 (2.6)	53
Extracted features (Video)	Action units occurrence duration	1 (6.3)	58
	Features based on facial movement and optical Flow	1 (6.3)	59
	Deviation between 68 Facial Landmarks	1 (6.3)	61
	Histogram of gradients feature	1 (6.3)	92
Extracted features (Audio)	Fundamental frequency	4 (25.0)	64, 65, 83,92
	Intensity	3 (18.8)	64, 83,92
	Mel-Frequency Cepstral Coefficients plus the 0th coefficient	3 (18.8)	64, 65,92
	N-gram features	3 (18.8)	67–69
	Formant frequencies and their bandwidths (F1, F2, F3, B1, B2, B3)	2 (12.5)	64,65
	Jitter	3 (18.8)	64, 65,104
	Shimmer	3 (18.8)	64, 65,104

Table 1 (continued) | Characteristics of all included studies (n = 80)

Category	Feature	Number of studies(%)	References
	Log energy	1 (6.3)	65
	Delta Mel-Frequency Cepstral Coefficients	1 (6.3)	65
	Acceleration Mel-Frequency Cepstral Coefficients	1 (6.3)	65
	Landmark duration features	1 (6.3)	69
	Spectrogram	1 (6.3)	83
	Teager energy operator	1 (6.3)	83
Extracted features (Eye Movement)	Fixation duration	6 (85.7)	71–75,87
	Fixation count	6 (85.7)	71–74,76,87
	Saccade amplitude	4 (57.1)	71–73,75
	Mean reaction time	3 (42.9)	72–74
	First fixation time	2 (28.6)	72,73
	Saccade lasting duration	2 (28.6)	71,75
	Saccade occurrence	1 (14.3)	71
	Saccade orientation	1 (14.3)	71
	Saccade speed	1 (14.3)	75
	Mean reaction speed	1 (14.3)	72
Extracted features (Gait)	Distance between joints	3 (50.0)	77, 78,82
	Speed of joints	3 (50.0)	77, 79,82
	Angle between joints	2 (33.3)	77,82
	Acceleration of joints	2 (33.3)	77,79
	Not reported	2 (33.3)	62,80
	Power spectral density	1 (16.7)	79
AI Algorithms	Support vector machine	36 (45.0)	27, 30, 31, 34, 35, 37, 38, 42, 45, 46, 49, 52, 53, 55, 57, 59–61, 63, 64, 66–68, 70–76, 79, 82, 86, 91, 103
	K-nearest neighbor	15 (18.8)	27, 30, 31, 35, 38, 40–42, 45, 55, 70, 75, 76, 82, 86
	Convolutional neural network	17 (21.3)	32, 33, 39, 51, 54–56, 62, 68, 69, 80, 87, 88, 90, 100, 101, 104
	Logistic regression	11 (13.8)	29–31, 35, 38, 42, 45, 70, 72, 81, 86
	Decision tree	10 (12.5)	27, 28, 30, 31, 38, 46, 58, 75, 76, 86
	Random Forest	10 (12.5)	35, 46, 50, 53, 70, 75, 79, 82, 86, 88
	Naïve bayes	5 (6.3)	30, 45, 70, 74, 86
	CNN + LSTM	5 (6.3)	36, 47, 65, 84, 85
	Extreme Gradient Boosting	3 (3.8)	31, 38, 50
	Gradient Boosting Decision Tree	3 (3.8)	53, 75, 79
	Long Short-Term Memory	4 (5.0)	62, 90, 102, 106
	Neural Network	1 (1.3)	30
	GNN + LSTM	1 (1.3)	83
	Graph Neural Network	1 (1.3)	43
	Graph Convolutional Network	1 (1.3)	44
	Adaptive Boosting	1 (1.3)	53
	LSTM+Transformer	2 (2.5)	89, 105
	CNN + LSTM+Transformer	1 (1.3)	92
	Transformer	1 (1.3)	98
Datasources	Closed	42 (52.5)	28, 29, 32, 34–37, 41, 42, 46–48, 50–53, 55–58, 60, 61, 63, 70–82, 86, 98
	Open	38 (47.5)	27, 30, 31, 33, 38–40, 43–45, 49, 54, 59, 62, 64–69, 83–85, 87–92, 99–106
Ground truth assessment	DSM-IV and -V	25 (31.3)	30, 35, 47, 64, 65, 67–70, 86, 87, 102–105
	PHQ-8 and -9	26 (32.5)	36, 37, 52, 53, 57, 60, 75, 79, 80, 91, 98
	Beck Depression Inventory-II	15 (18.8)	34, 37, 42, 58, 59, 61

Table 1 (continued) | Characteristics of all included studies (n = 80)

Category	Feature	Number of studies(%)	References
	17-item Hamilton Depression Rating Scale	11 (13.8)	37, 42,50
	Mini International Neuropsychiatric Interview	6 (7.5)	37,42
	Life Event Scale	3 (3.8)	100, 101,106
	Childhood Trauma Questionnaire	2 (2.5)	41, 46, 51, 56, 63,71–73,92
	Self-Rating Depression Scale	3 (3.8)	27, 28, 30, 31, 38, 46, 58, 75, 76,86
	Not reported	9 (11.3)	35, 46, 50, 53, 70, 75, 79, 82, 86,88
Task	Eye closed (EEG)	24 (63.1)	27–29,31, 33,36–46,48, 49,52–54,56,82–85
	Emotional Image Stimulation Task(EEG)	18 (47.4)	31,33–35,37–40,43, 44, 47, 66,83–86,98,99
	Eye open(EEG)	10 (26.3)	27, 28, 32, 41, 45, 46, 49, 50, 54,56
	Oddball task(EEG)	3 (7.9)	45, 54,55
	Clinical Interview(Audio)	23 (76.7)	62–65,67–69,82–85,87–92,98–100,106
	Reading Task(Audio)	5 (16.7)	63–65,67–69,87,102–105
	Six elicitation tasks(Audio)	1 (6.3)	66
	Free viewing(Eye Movement)	7 (100)	70–75,86
	Frame tracking(Eye Movement)	1 (14.3)	71
	Clinical Interview(Video)	8 (50.0)	57,87–90,92, 98,101
	Emotional Image Stimulation Task(Video)	6 (37.5)	58–61,91,99
	Free Walking(Gait)	4 (66.7)	76–78,81
	Posture Task(Gait)	2 (33.3)	79,80
	Not reported	2 (2.5)	30,51

some EEG research findings. Several EEG studies have identified a link between hemispheric asymmetry in the cortical frontal regions and depressive symptoms¹⁰⁸. The current research on left and right frontal lobe electrodes consistently shows that, during eye-open/eye-closed task^{29,41,56}, the EEG collected by electrodes in the right frontal lobe contains more depression-related information than the electrodes in the left frontal lobe. This suggests that patients with depression may exhibit abnormally increased right frontal activity at rest, providing important insights for future research in this area¹⁰⁹. In addition, there are two main approaches for audio-based depression screening: extracting acoustic features for traditional machine learning or using end-to-end deep learning on raw timing audio signals or spectrograms. Our subgroup meta-analysis showed that machine learning with prosodic features achieved similar classification performance to deep learning methods (AUC = 0.92 and 0.94), with 17% higher specificity but 8% lower sensitivity (Fig. 3). However, the reduction in sensitivity increases the risk of missing depression cases. Therefore, achieving an optimal balance between sensitivity and specificity is crucial for developing effective depression screening tools, as is the case for all modalities. Despite the promise of EEG and audio signals, other single modalities have not performed as well, leading to the exploration of multi-modal approaches.

For screening methods using uni-modal information, feature extraction-based algorithms outperformed other algorithms in subgroup analyses (Table 3). This discrepancy may be attributed to the rapid advancements in the field of feature extraction, particularly concerning EEG¹⁰⁷ and voice data¹⁰⁸. In this study, we compiled statistics on the frequency and selection rates of specific features used across modalities based on the literature (Table 1). Through this statistical analysis, we can identify frequently used features, such as power spectral density and Lempel–Ziv complexity from EEG data, and fixation duration from eye movement data, and assess their importance in both single-modality and multi-modal

screening. This provides valuable reference data for optimizing feature selection strategies and enhancing the accuracy of depression screening in future research.

There are two main multi-modal fusion methods currently prevalent in current research. The first method combines EEG and audio, which discriminates depression from NC with a pooled SE of 0.89 (95% CI: 0.83–0.92), a pooled SP of 0.88 (95% CI: 0.85–0.91), and an AUC of 0.93 (95% CI: 0.91–0.95). The second method integrates video, audio, and text, which discriminates depression from NC with a pooled SE of 0.88 (95% CI: 0.84–0.92), a pooled SP of 0.91 (95% CI: 0.88–0.93), and an AUC of 0.95 (95% CI: 0.93–0.97) (Table 3). This observation underscores that AI-based multi-modal fusion is a promising tool for screening depression. The improvements observed may be attributed to several factors: AI-assisted multi-modal physiological and behavioral information can effectively integrate diverse data formats, thereby reducing the noise and variability inherent in uni-modal data. In addition, the standardized and stable screening process of AI can minimize subjective errors. Moreover, while patients might try to conceal their conditions during a clinical evaluation, they cannot easily hide physiological signals (e.g., EEG) and behavioral reactions (e.g., micro-expressions).

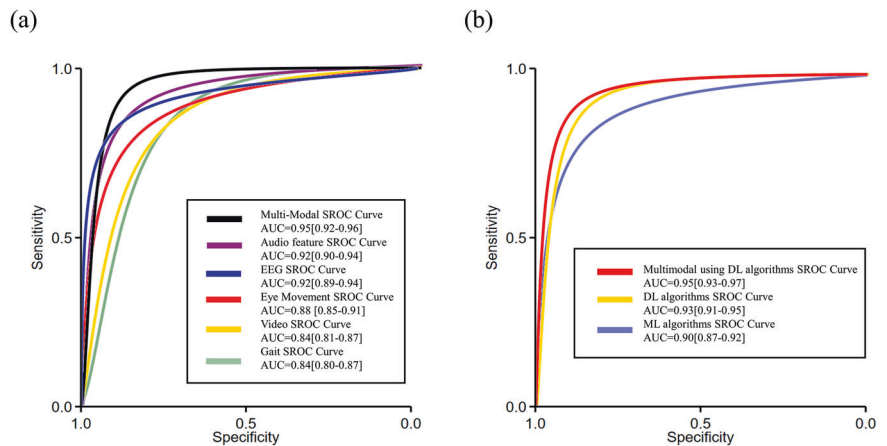
Despite the progress, there are still several technical improvements needed in current research. Many studies on ML methods continue to rely on classic models such as SVM and KNN, which primarily concatenate features from different modalities. Exploring more advanced models could be beneficial. For instance, Extreme Gradient Boosting (XGBoost) and its derivatives (e.g., LightGBM, CatBoost) not only often outperform traditional classifiers in various tasks but also offer visualization of feature importance scores or decision trees, enhancing interpretability in multi-modal data analysis^{110–112}. This transparency is crucial, as understanding the contribution of each modality can lead to better model interpretation and refinement. Multi-kernel learning allows the integration of multiple

Table 2 | Pooled estimates and meta-regression of pooled performance for depression diagnosis across modalities

Parameter	No. of tables	AUC (95% CI)	Sensitivity		p value ^a	Specificity		p value ^a	LR+ (95% CI)	LR- (95% CI)
			SE (95% CI)	I2 (95% CI)		SP (95% CI)	I2 (95% CI)			
Single modal										
EEG										
Overall	88	0.92 (0.89–0.94)	0.85 (0.84–0.87)	32.64 (14.76–50.51)		0.86 (0.84–0.88)	56.14 (45.67–66.61)		6.03 (5.21–6.97)	0.17 (0.15–0.19)
Highest performance	33	0.94 (0.92–0.96)	0.86 (0.83–0.89)	32.72 (2.59–62.84)		0.89 (0.85–0.92)	63.42 (49.11–77.72)		8.30 (6.46–10.66)	0.15 (0.12–0.18)
Eye movement										
Overall	24	0.88 (0.85–0.91)	0.77 (0.72–0.81)	33.15 (0.00–66.40)		0.86 (0.83–0.89)	47.67 (22.76–72.57)		5.61 (4.38–7.19)	0.27 (0.22–0.33)
Highest performance	6	0.90 (0.87–0.92)	0.81 (0.74–0.87)	77.49 (59.55–95.44)		0.85 (0.77–0.90)	90.00 (83.56–96.45)		5.32 (3.27–8.64)	0.22 (0.15–0.33)
Video										
Overall	17	0.86 (0.82–0.88)	0.83 (0.78–0.87)	81.12 (72.84–89.40)		0.75 (0.70–0.80)	75.92 (64.65–87.20)		3.24 (2.67–4.18)	0.23 (0.17–0.31)
Highest performance	6	0.87 (0.84–0.90)	0.82 (0.72–0.89)	81.10 (66.67–95.54)		0.78 (0.66–0.87)	83.23 (70.81–95.66)		3.71 (2.13–6.44)	0.23 (0.12–0.41)
Audio										
Overall	28	0.92 (0.90–0.94)	0.86 (0.81–0.89)	79.56 (72.48–86.65)		0.86 (0.81–0.90)	89.29 (86.21–92.38)		6.13 (4.55–8.27)	0.17 (0.12–0.23)
Highest performance	16	0.95 (0.92–0.96)	0.86 (0.80–0.91)	81.18 (72.67–89.69)		0.90 (0.86–0.93)	84.24 (77.46–91.03)		8.96 (6.09–13.17)	0.15 (0.10–0.22)
Gait										
Overall	25	0.84 (0.80–0.87)	0.79 (0.73–0.84)	91.67 (89.3–94.03)		0.77 (0.73–0.81)	83.23 (77.40–89.06)		3.45 (2.78–4.29)	0.28 (0.21–0.387)
Highest performance	6	0.90 (0.87–0.92)	0.85 (0.75–0.91)	85.64 (75.42–95.86)		0.83 (0.78–0.88)	71.85 (48.22–95.48)		5.08 (3.48–7.42)	0.19 (0.11–0.32)
Multi-modal										
Overall	39	0.95 (0.92–0.96)	0.88 (0.85–0.91)	66.67 (55.53–77.82)		0.90 (0.88–0.91)	50.18 (31.74–68.61)		8.67 (7.14–10.53)	0.13 (0.10–0.17)
Highest performance	15	0.96 (0.94–0.97)	0.89 (0.84–0.93)	70.63 (55.20–86.07)		0.91 (0.88–0.94)	63.37 (43.04–83.70)		10.09 (7.18–14.18)	0.12 (0.08–0.18)
Algorithm										
Deep learning	64	0.93 (0.91–0.95)	0.84 (0.80–0.88)	88.62 (86.08–91.16)	<0.001	0.89 (0.87–0.91)	85.24 (81.69–88.79)	<0.001	7.56 (6.09–9.39)	0.18 (0.14–0.23)
Machine learning	157	0.90 (0.87–0.92)	0.83 (0.82–0.85)	64.83 (58.90–70.76)		0.82 (0.81–0.84)	74.33 (70.36–78.30)		4.77 (4.32–5.26)	0.20 (0.18–0.22)
Multi-modal with Deep Learning	20	0.95 (0.93–0.97)	0.87 (0.83–0.90)	57.85 (36.94–78.77)		0.91 (0.88–0.93)	61.97 (43.56–80.39)		9.27 (6.97–12.33)	0.15 (0.11–0.20)

^ap value for heterogeneity between subgroups with meta-regression analysis.

Fig. 3 | Hierarchical SROC curves of studies in meta-analysis of depression and NC classification using different paradigms. a The SROC curves comparing uni-modal and multi-modal approaches. **b** The SROC curves stratified by deep learning and machine learning methods.



kernels¹¹³, each tailored to a specific modality, to better synthesize heterogeneous data¹¹⁴. This approach captures the unique features of each modality more effectively, thereby improving overall model performance.

Furthermore, methods using DL have improved overall screening accuracy by 3% compared to those using traditional ML methods (Table 2). Despite this improvement, current research still faces several challenges. For instance, due to the significant differences in data formats between input modalities, 89% of the multi-modal DL investigations included employ a late fusion strategy ($n = 9$). This strategy involves concatenating features at the decision-making layer. While late fusion requires less data and allows for easier integration of multiple sensing strategies and algorithms, it may not fully leverage the complementary strengths of different modalities¹¹⁵. Additionally, this approach is more susceptible to overfitting, especially with small datasets¹¹⁶. To address these limitations, we suggest exploring more sophisticated fusion and alignment techniques in multi-modal DL screening model, especially in small sample size¹¹⁷.

For future work in multi-modal AI depression screening, several areas need focus. Integrating advanced DL techniques is crucial, such as implementing cross-modal attention mechanisms, which allow models to dynamically concentrate on the most relevant features of each information¹¹⁷ (Fig. 4). Additionally, techniques like cross-task and cross-dataset transfer learning could be used to help enhance the generalization capability of models, ensuring robustness and stability in practical applications¹¹⁸. Moreover, integrating multi-modal frameworks with large language models (LLMs) may help leverage rich contextual semantics, leading to more personalized depression screening. Developing more tools for interpretability is also essential for understanding and improving AI systems. For instance, utilizing methods such as SHAP (Shapley Additive Explanations) values can help clinicians grasp the reasoning behind ML model decisions, thereby increasing trust in screening outcomes¹¹⁹.

In addition to the model framework, proper standardization and preprocessing of data are also important to remove noise and format differences between datasets and modalities^{120,121}. For instance, in this study, all literature involving EEG preprocessed the data to remove artifacts such as EMG, ECG, and frequency signals outside the EEG band, including power frequency signals. Similarly, due to factors like aliasing, high-order harmonic distortion, and high frequencies from the human vocal apparatus and audio collection equipment, audio signals require preprocessing such as pre-emphasis, framing, and windowing. Video-based studies also performed face detection and alignment between adjacent frames, with some extracting 68 facial landmarks and action units for further feature extraction. Finally, gait-based studies using Kinect devices processed the three-dimensional coordinate stream of 25 joints generated by the Kinect. This involved discarding incorrectly estimated coordinates or frames with deformed skeletons and performing coordinate transformations. Most studies in this review have considered these steps; however, the lack of a unified standard for preprocessing certain modalities (e.g., gait, eye movement) may lead to

bias. Future research should focus on developing more robust and automated preprocessing pipelines to efficiently handle various formats.

Despite the promising performance of existing models, the challenge of data availability remains significant. Depression data is sensitive, making it difficult to collect diverse datasets for estimating the severity of depression. Among the studies included in this meta-analysis (the dataset used in each study is listed in Supplementary Table 1), six public datasets are highlighted: the PRED + CT dataset (providing EEG data)¹²²; the AVEC 2011–2021 datasets (e.g., DAIC-WOZ and E-DAIC, featuring video, audio, and text modalities mainly in English)^{123–130}; the MODMA dataset (offering EEG and audio modalities in Chinese)¹³¹; the BlackDog database (providing video and audio modalities in English); the CMDC dataset (including video, audio, and text modalities in Chinese)⁹⁸; and the EATD-Corpus dataset (focusing on audio modalities in Chinese)¹⁰⁶. However, each of these public datasets, as well as the private datasets used in included studies, contains no more than 280 subjects (with sample sizes ranging from 24 to 275), which hampers the ability to identify clinical biomarkers universally and reliably, especially with DL approaches^{132,133}. We advocate for the establishment of a larger, more diverse, and standardized database of depressed patients, with multi-modal data collected under consistent conditions, equipment, and configurations.

While data-driven depression screening using modalities such as EEG, eye movements, facial expressions, speech, and gait has the potential to relieve clinical workloads and enable early intervention, it also raises ethical challenges. There is a risk of misuse by third parties, such as employers or insurers, especially if data are used without explicit patient consent or understanding. Researchers and clinicians must ensure transparent communication regarding data use, secure robust data protection measures, and guarantee that participation is voluntary and withdrawable at any time. Continuous ethical oversight is essential to safeguard individuals' rights and promote the responsible application of these technologies¹³⁴.

This study has several notable limitations. Firstly, we focused on English-language articles, which may overlook valuable findings from non-English studies. Secondly, as most pooled estimates for the algorithms were based on a limited number of studies and limited sample sizes within these studies, the results may be subject to bias. Additionally, the subgroup analysis in this study still exhibited certain heterogeneity. Further research should examine the classification performance of AI algorithms in more homogeneous patient groups. Moreover, few of the studies involved cross-ethnic or cross-age research, making it challenging to verify whether models trained on one specific ethnic group or age group are equally effective for others. Additionally, we excluded studies that only reported continuous metrics such as MAE or RMSE to maintain analytic consistency, as such assessments have not been conducted for all modalities in the existing literature. Future research could further explore the use of continuous outcome measures to broaden current insights. Some relevant studies, such as Audiface¹³⁵ and Audibert⁵, which only reported F1 scores without

Table 3 | Summary performance of subgroup analysis of depression diagnosis across modalities

Parameter	No. of tables	AUC (95% CI)	Sensitivity		p value ^a	Specificity		p value ^a
			SE (95% CI)	I2 (95% CI)		SP (95% CI)	I2 (95% CI)	
EEG								
Overall	88	0.92 (0.89–0.94)	0.85 (0.84–0.87)	32.64 (14.76–50.51)		0.86 (0.84–0.88)	56.14 (45.67–66.61)	
Feature								
Timing EEG signals	23	0.91 (0.88–0.93)	0.85 (0.80–0.88)	0.86 (0.00–63.87)		0.84 (0.79–0.88)	51.47 (27.70–75.74)	
Linear and non-linear features	42	0.92 (0.89–0.94)	0.85 (0.83–0.88)	45.72 (25.97–65.47)		0.86 (0.82–0.88)	59.44 (45.69–73.19)	
Graph method	20	0.93 (0.90–0.95)	0.85 (0.81–0.89)	31.64 (00.00–68.76)		0.87 (0.82–0.91)	58.06 (37.28–78.85)	
Gait								
Overall	25	0.84 (0.81–0.87)	0.79 (0.73–0.84)	91.67 (89.3–94.03)		0.84 (0.81–0.87)	83.23 (77.40–89.06)	
Feature								
Skeleton joints coordinate stream	5	0.67 (0.63–0.71)	0.50 (0.44–0.56)	37.96 (0.00–99.37)		0.73 (0.68–0.77)	24.14 (0.00–92.42)	
Linear gait feature	12	0.91 (0.88–0.93)	0.87 (0.83–0.90)	75.73 (62.09–89.37)		0.84 (0.81–0.87)	49.15 (15.37–82.94)	
Silhouette feature	8	0.79 (0.75–0.83)	0.77 (0.74–0.80)	0.00 (0.00–100.00)		0.66 (0.63–0.70)	30.68 (0.00–86.71)	
Audio								
Overall	28	0.92 (0.90–0.94)	0.86 (0.81–0.89)	79.56 (72.48–86.65)		0.86 (0.81–0.90)	89.29 (86.21–92.38)	
Feature								
Timing Audio Signal	15	0.94 (0.92–0.96)	0.84 (0.78–0.88)	80.13 (72.01–88.26)		0.92 (0.88–0.94)	90.13 (86.83–93.44)	
Prosodic features	6	0.92 (0.89–0.94)	0.92 (0.88–0.95)	0.00 (0.00–100.00)		0.75 (0.69–0.80)	0.00 (0.00–100.00)	
Bigram Features	4	/	/	/		/	/	
Features Extracted by N-gram	3	/	/	/		/	/	
Video								
Overall	17	0.84 (0.80–0.87)	0.82 (0.77–0.86)	80.59 (71.73–89.44)		0.74 (0.70–0.78)	70.44 (55.38–85.50)	
Feature								
Timing video data	3	/	/	/		/	/	
Linear feature based on action units	3	/	/	/		/	/	
Extracted video frame	8	0.88 (0.85–0.91)	0.85 (0.81–0.89)	68.54 (45.27–91.80)		0.78 (0.74–0.81)	64.87 (38.27–91.48)	
Linear feature based on 68 facial landmarks	2	/	/	/		/	/	
Multi-modal								
Overall	39	0.95 (0.92–0.96)	0.88 (0.85–0.91)	66.67 (55.53–77.82)		0.90 (0.88–0.91)	50.18 (31.74–68.61)	
Modal								
EEG and audio	19	0.93 (0.91–0.95)	0.89 (0.83–0.92)	69.53 (55.20–83.86)		0.88 (0.85–0.91)	10.65 (0.00–56.36)	
Video and audio	20	0.95 (0.93–0.97)	0.88 (0.84–0.92)	64.68 (47.88–81.47)		0.91 (0.88–0.93)	65.78 (49.63–81.92)	

^ap value for heterogeneity between subgroups with meta-regression analysis.

providing sufficient data (e.g., TP, TN, FP, FN) required for meta-analysis, could not be included, which may have introduced inclusion bias. Finally, potentially due to the influence of public datasets, most of the multi-modal fusion methods in the literature focused on two modalities, preventing further investigation into the effectiveness of each modality.

This study highlights the significant potential of artificial intelligence algorithms in multi-modal depression screening and outlines the promising prospects and development directions for each modality. Despite challenges such as the risk of false positives and false negatives, data privacy and security concerns, and regulatory approval requirements, AI remains a

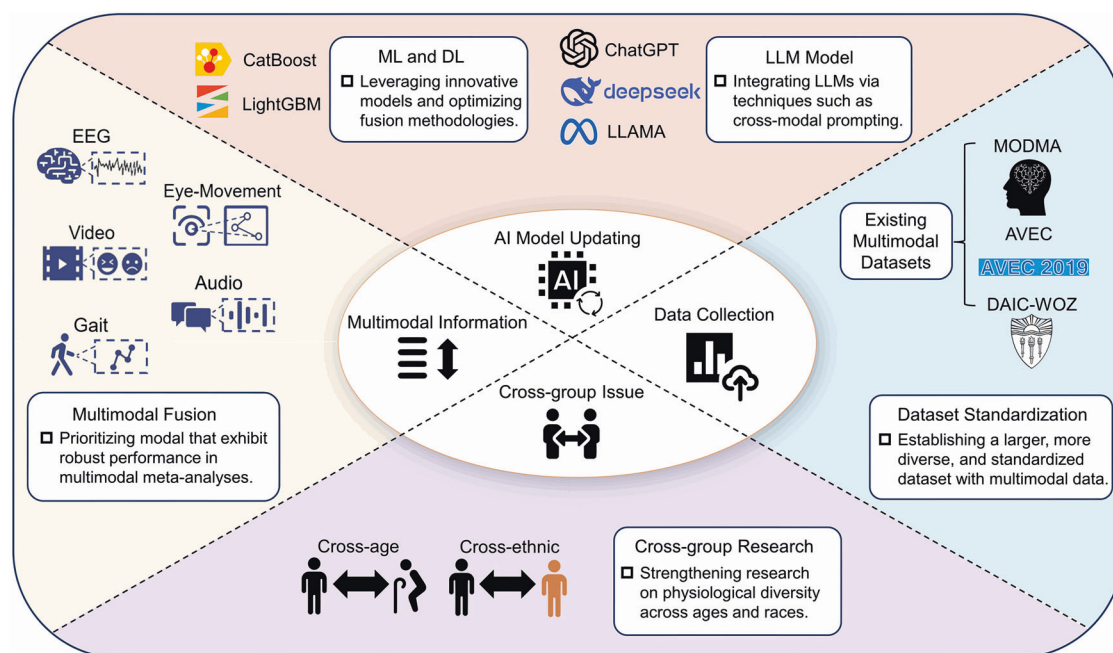


Fig. 4 | Outlook for multi-modal physiological and behavioral information integration. Illustration of Multi-modal Fusion, AI Model Updating, Dataset Standardization, and Cross-group Research in Diverse Applications.

valuable tool for assisting doctors with diagnosis. Additionally, we emphasize the need to enhance the study design of AI-based depression screening systems and advocate for the creation of data-sufficient standardized databases to support further research.

Methods

The systematic review and meta-analysis were performed according to the standard PRISMA (preferred reporting items for systematic reviews and meta-analyses)¹³⁶ (Supplementary Table 7). The study was registered in the PROSPERO (CRD420251049107).

Search strategy and eligibility criteria

Two researchers with experience in computer science and biomedical engineering independently conducted literature searches and records. The agreement was then achieved by a third reviewer with the expertise of data analysis. All these searches were performed on Google Scholar, Web of Science Core Collection, and IEEE Xplore database for publications up to June 7, 2025. Only English-language articles were included. To investigate the strategies of automated clinically informative depression screening with both single and multiple modalities, we conducted a literature search using a combination of key terms from three categories: (1) depression-related disorders (“depression” OR “major depressive disorder” OR “mood disorder”), (2) screening processes (“detection” OR “diagnosis” OR “screening”), and (3) modalities (“EEG” OR “eye movement” OR “gait” OR “speech” OR “vocal” OR “audio” OR “audiovisual” OR “facial” OR “visual” OR “video” OR “multi-modal”). Search queries were constructed by combining at least one term from each category with the AND operator, while terms within each category were combined using OR. First pre-screen by looking at abstracts and titles to filter out irrelevant articles. Eligibility is then checked against our criteria (Fig. 2) and unqualified full-text review entries are removed. Eligible studies that reported AI-assisted each modality for the screening of Depression with classification outcomes such as sensitivity (SE) and specificity (SP) were then used to calculate the 2 × 2 contingency tables.

Studies were included if they met the following criteria: (1) the main focus was on depression (excluding studies centered on other disorders such as schizophrenia, Alzheimer’s disease, brain trauma, or cancer); (2) only technical research was included (excluding reviews and non-technical papers); (3) the study purposed and evaluated an approach for depression

diagnosis that used machine learning or deep learning models to classify depression; (4) comprehensive classification performance results were reported, specifically accuracy, sensitivity, specificity, recall, precision, or an explicit confusion matrix, enabling direct extraction or calculation of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

Data analysis

Details of papers for quality review were manually summarized in a spreadsheet, including title, modality, year of publication, source of data, type of patient’s depression, number of patients and normal control, types of features, algorithm model, and classification performance data.

Classification performance data for included studies were extracted, and contingency tables were created. These data included accuracy, sensitivity, specificity, recall, precision, or an explicit confusion matrix, which enabled direct extraction or calculation of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Contingency tables were used to construct SROC curves and forest plots, as well as to calculate pooled sensitivity and specificity. If a study provided multiple contingency tables for different AI algorithms, the contingency tables for different AI algorithms were used independently.

Risk of bias and applicability appraisal

The risk of bias and applicability of all selected studies were assessed by using the QUADAS-2 criteria¹³⁷. It provides researchers with a standardized framework for assessing the risk of bias and applicability when conducting reviews that evaluate the accuracy of AI-assisted screening tests.

Data synthesis

We performed a meta-analysis of studies using contingency tables to estimate the classification performance of AI methods, including machine learning (ML) or DL algorithms. As differences between studies were assumed, a random effects model was performed. We conducted a meta-analysis of all single-modal and multi-modal research literature data; number of research literature for all modalities was greater than 5, which meets the requirements of random-effect analysis¹³⁸. We used contingency tables to construct stratified summary receiver operating characteristic (SROC) curves and calculated summary sensitivity and specificity for

predicting high heterogeneity. The combined curve is the corresponding 95% confidence region and 95% prediction region plotted around the average estimate of SE, SP, and AUC in the SROC plot. Heterogeneity was assessed, using the I² statistic. Meanwhile, all SROC curves were summarized on one graph to facilitate comparison and calculated using STATA statistical software (version 17.0) (Midas and Metandi modules; StataCorp). Statistical significance is shown when the $P < 0.05$.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

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References

1. Diseases, G. B. D. & Injuries, C. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet* **396**, 1204–1222 (2020).
2. Mitchell, A. J., Vaze, A. & Rao, S. Clinical diagnosis of depression in primary care: a meta-analysis. *Lancet* **374**, 609–619 (2009).
3. de Aguiar Neto, F. S. & Rosa, J. L. G. Depression biomarkers using non-invasive EEG: A review. *Neurosci. Biobehav. Rev.* **105**, 83–93 (2019).
4. Stolicyn, A., Steele, J. D. & Series, P. Prediction of depression symptoms in individual subjects with face and eye movement tracking. *Psychol. Med.* **52**, 1784–1792 (2022).
5. Toto, E., Tlachac, M. & Rundensteiner, E. A. AudiBERT: a deep transfer learning multimodal classification framework for depression screening. In *Proceedings 30th ACM International Conference on Information & Knowledge Management* 4145–4154. <https://doi.org/10.1145/3459637.3481895> (2021).
6. Francese, R. & Attanasio, P. Emotion detection for supporting depression screening. *Multimed. Tools Appl.* **82**, 12771–12795 (2023).
7. Hasler, G. Pathophysiology of depression: do we have any solid evidence of interest to clinicians?. *World Psychiatry* **9**, 155 (2010).
8. Han, S. et al. Orbitofrontal cortex-hippocampus potentiation mediates relief for depression: a randomized double-blind trial and TMS-EEG study. *Cell Rep. Med.* **4**, 101060 (2023).
9. Rajkowska, G. et al. Morphometric evidence for neuronal and glial prefrontal cell pathology in major depression. *Biol. Psychiatry* **45**, 1085–1098 (1999).
10. Drevets, W. C. et al. Subgenual prefrontal cortex abnormalities in mood disorders. *Nature* **386**, 824–827 (1997).
11. Mayberg, H. S. et al. Reciprocal limbic-cortical function and negative mood: converging PET findings in depression and normal sadness. *Am. J. Psychiatry* **156**, 675–682 (1999).
12. Esterman, M. et al. Frontal eye field involvement in sustaining visual attention: evidence from transcranial magnetic stimulation. *Neuroimage* **111**, 542–548 (2015).
13. Corbetta, M. & Shulman, G. L. Control of goal-directed and stimulus-driven attention in the brain. *Nat. Rev. Neurosci.* **3**, 201–215 (2002).
14. Lazarov, A., Ben-Zion, Z., Shamaï, D., Pine, D. S. & Bar-Haim, Y. Free viewing of sad and happy faces in depression: a potential target for attention bias modification. *J. Affect. Disord.* **238**, 94–100 (2018).
15. Girard, J. M. et al. Nonverbal social withdrawal in depression: evidence from manual and automatic analysis. *Image Vis. Comput.* **32**, 641–647 (2014).
16. Deligianni, F., Guo, Y. & Yang, G. Z. From emotions to mood disorders: a survey on gait analysis methodology. *IEEE J. Biomed. Health Inf.* **23**, 2302–2316 (2019).
17. Kebets, V. et al. Somatosensory-motor dysconnectivity spans multiple transdiagnostic dimensions of psychopathology. *Biol. Psychiatry* **86**, 779–791 (2019).
18. Schrijvers, D., Hulstijn, W. & Sabbe, B. G. Psychomotor symptoms in depression: a diagnostic, pathophysiological and therapeutic tool. *J. Affect. Disord.* **109**, 1–20 (2008).
19. Sobin, C. & Sackeim, H. A. Psychomotor symptoms of depression. *Am. J. Psychiatry* **154**, 4–17 (1997).
20. He, L. et al. Deep learning for depression recognition with audiovisual cues: a review. *Inf. Fusion* **80**, 56–86 (2022).
21. Abd-Alrazaq, A. et al. Systematic review and meta-analysis of performance of wearable artificial intelligence in detecting and predicting depression. *npj Digit. Med.* **6**, 84 (2023).
22. Mangalik, S. et al. Robust language-based mental health assessments in time and space through social media. *npj Digit. Med.* **7**, 109 (2024).
23. Kaczmarczyk, R., Wilhelm, T. I., Martin, R. & Roos, J. Evaluating multimodal AI in medical diagnostics. *npj Digit. Med.* **7**, 205 (2024).
24. Kline, A. et al. Multimodal machine learning in precision health: a scoping review. *npj Digit. Med.* **5**, 171 (2022).
25. Cunningham, S., Hudson, C. C. & Harkness, K. Social media and depression symptoms: a meta-analysis. *Res. Child Adolesc. Psychopathol.* **49**, 241–253 (2021).
26. Yoon, S., Kleinman, M., Mertz, J. & Brannick, M. Is social network site usage related to depression? A meta-analysis of Facebook–depression relations. *J. Affect. Disord.* **248**, 65–72 (2019).
27. Yang, J. et al. Cross-subject classification of depression by using multiparadigm EEG feature fusion. *Comput. Methods Prog. Biomed.* **233**, 107360 (2023).
28. Mohammadi, M. et al. Data mining EEG signals in depression for their diagnostic value. *BMC Med. Inform. Decis. Mak.* **15**, 1–14 (2015).
29. Koo, P. C. et al. Combined cognitive, psychomotor and electrophysiological biomarkers in major depressive disorder. *Eur. Arch. Psychiatry Clin. Neurosci.* **269**, 823–832 (2019).
30. Nassibi, A., Papavassiliou, C. & Atashzar, S. F. Depression diagnosis using machine intelligence based on spatio-spectrotemporal analysis of multi-channel EEG. *Med. Biol. Eng. Comput.* **60**, 3187–3202 (2022).
31. Soni, S., Seal, A., Yazidi, A. & Krejcar, O. Graphical representation learning-based approach for automatic classification of electroencephalogram signals in depression. *Comput. Biol. Med.* **145**, 105420 (2022).
32. Seal, A. et al. DeprNet: a deep convolution neural network framework for detecting depression using EEG. *IEEE Trans. Instrum. Meas.* **70**, 1–13 (2021).
33. Liu, B., Chang, H., Peng, K. & Wang, X. An end-to-end depression recognition method based on EEGNet. *Front. Psychiatry* **13**, 864393 (2022).
34. Li, X. et al. Depression recognition using machine learning methods with different feature generation strategies. *Artif. Intell. Med.* **99**, 101696 (2019).
35. Li, X., Hu, B., Sun, S. & Cai, H. EEG-based mild depressive detection using feature selection methods and classifiers. *Comput. Methods Prog. Biomed.* **136**, 151–161 (2016).
36. Song, X., Yan, D., Zhao, L. & Yang, L. LSDD-EEGNet: an efficient end-to-end framework for EEG-based depression detection. *Biomed. Signal Process. Control* **75**, 103612 (2022).
37. Shen, J., Zhang, X., Wang, G., Ding, Z. & Hu, B. An improved empirical mode decomposition of electroencephalogram signals for depression detection. *IEEE Trans. Affect. Comput.* **13**, 262–271 (2019).
38. Soni, S., Seal, A., Mohanty, S. K. & Sakurai, K. Electroencephalography signals-based sparse networks integration using a fuzzy ensemble technique for depression detection. *Biomed. Signal Process. Control* **85**, 104873 (2023).
39. Jiang, W. et al. EEG-based subject-independent depression detection using dynamic convolution and feature adaptation. In *Proceedings International Conference on Swarm Intelligence* 272–283 (2023).

40. Tasci, G. et al. Automated accurate detection of depression using twin Pascal's triangles lattice pattern with EEG Signals. *Knowl. Based Syst.* **260**, 110190 (2023).
41. Sadiq, M. T., Akbari, H., Siuly, S., Yousaf, A. & Rehman, A. U. A novel computer-aided diagnosis framework for EEG-based identification of neural diseases. *Comput. Biol. Med.* **138**, 104922 (2021).
42. Shi, Q. et al. Depression detection using resting state three-channel EEG signal. Preprint at *arXiv* <https://doi.org/10.48550/arXiv.2002.09175> (2020).
43. Chen, T., Guo, Y., Hao, S. & Hong, R. Exploring self-attention graph pooling with EEG-based topological structure and soft label for depression detection. *IEEE Trans. Affect. Comput.* **13**, 2106–2118 (2022).
44. Wang, H.-G., Meng, Q.-H., Jin, L.-C., Wang, J.-B. & Hou, H.-R. AMG: a depression detection model with autoencoder and multi-Head graph convolutional network. In *Proceedings 2023 42nd Chinese Control Conference (CCC)* 8551–8556 (2023).
45. Shivcharan, M., Bobby, K. & Sridevi, V. EEG based machine learning models for automated depression detection. In *Proceedings 2023 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)* 1–6 (2023).
46. Bai, R., Guo, Y., Tan, X., Feng, L. & Xie, H. An EEG-based depression detection method using machine learning model. *Int. J. Pharma Med. Biol. Sci.* **10**, 17–22 (2021).
47. Fan, Y., Yu, R., Li, J., Zhu, J. & Li, X. EEG-based mild depression recognition using multi-kernel convolutional and spatial-temporal feature. In *Proceedings 2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 1777–1784 (2020).
48. Tian, F. et al. The three-lead EEG sensor: introducing an EEG-assisted depression diagnosis system based on ant lion optimization. *IEEE Trans. Biomed. Circuits Syst.* **17**, 1305–1318 (2023).
49. Zhu, G. et al. Detecting depression using single-channel EEG and graph methods. *Mathematics* **10**, 4177 (2022).
50. Zhang, B. et al. Feature-level fusion based on spatial-temporal of pervasive EEG for depression recognition. *Comput. Methods Prog. Biomed.* **226**, 107113 (2022).
51. Li, D., Tang, J., Deng, Y. & Yang, L. Classification of resting state EEG data in patients with depression. In *Proceedings 2020 IEEE International Conference on E-health Networking, Application & Services (HEALTHCOM)* 1–2 (2020).
52. Mahato, S., Goyal, N., Ram, D. & Paul, S. Detection of depression and scaling of severity using six channel EEG data. *J. Med. Syst.* **44**, 1–12 (2020).
53. Lin, H. et al. MDD-TSVM: a novel semisupervised-based method for major depressive disorder detection using electroencephalogram signals. *Comput. Biol. Med.* **140**, 105039 (2022).
54. Rafiei, A., Zahedifar, R., Sitaula, C. & Marzbanrad, F. Automated detection of major depressive disorder with EEG signals: a time series classification using deep learning. *IEEE Access* **10**, 73804–73817 (2022).
55. Duan, L. et al. Machine learning approaches for MDD detection and emotion decoding using EEG signals. *Front. Hum. Neurosci.* **14**, 284 (2020).
56. Acharya, U. R. et al. Automated EEG-based screening of depression using deep convolutional neural network. *Comput. Methods Prog. Biomed.* **161**, 103–113 (2018).
57. Cohn, J. F. et al. Detecting depression from facial actions and vocal prosody. In *Proceedings 2009 3rd International Conference on Affective Computing and Intelligent Interaction and Workshops* 1–7 (2009).
58. Yang, M. et al. Undisturbed mental state assessment in the 5G era: a case study of depression detection based on facial expressions. *IEEE Wirel. Commun.* **28**, 46–53 (2021).
59. Shangquan, Z. et al. Dual-stream multiple instance learning for depression detection with facial expression videos. *IEEE Trans. Neural Syst. Rehabil. Eng.* **31**, 554–563 (2022).
60. Wang, Q., Yang, H. & Yu, Y. Facial expression video analysis for depression detection in Chinese patients. *J. Vis. Commun. Image Represent.* **57**, 228–233 (2018).
61. Hu, B., Tao, Y. & Yang, M. Detecting depression based on facial cues elicited by emotional stimuli in video. *Comput. Biol. Med.* **165**, 107457 (2023).
62. Lin, L., Chen, X., Shen, Y. & Zhang, L. Towards automatic depression detection: a BiLSTM/1D CNN-based model. *Appl. Sci.* **10**, 8701 (2020).
63. Kiss, G. & Vicsi, K. Comparison of read and spontaneous speech in case of automatic detection of depression. In *Proceedings 2017 8th IEEE International Conference on Cognitive Infocommunications (CogInfoCom)* 000213–000218 (2017).
64. Mobram, S. & Vali, M. Depression detection based on linear and nonlinear speech features in I-vector/SVDA framework. *Comput. Biol. Med.* **149**, 105926 (2022).
65. Vázquez-Romero, A. & Gallardo-Antolín, A. Automatic detection of depression in speech using ensemble convolutional neural networks. *Entropy* **22**, 688 (2020).
66. Huang, Z., Epps, J. & Joachim, D. Speech landmark bigrams for depression detection from naturalistic smartphone speech. In *Proceedings ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 5856–5860 (2019).
67. Huang, Z., Epps, J., Joachim, D. & Sethu, V. Natural language processing methods for acoustic and landmark event-based features in speech-based depression detection. *IEEE J. Sel. Top. Signal Process.* **14**, 435–448 (2019).
68. Huang, Z., Epps, J. & Joachim, D. Investigation of speech landmark patterns for depression detection. *IEEE Trans. Affect. Comput.* **13**, 666–679 (2019).
69. Yang, W. et al. Attention guided learnable time-domain filterbanks for speech depression detection. *Neural Netw.* **165**, 135–149 (2023).
70. Li, X., Cao, T., Sun, S., Hu, B. & Ratcliffe, M. Classification study on eye movement data: towards a new approach in depression detection. In *Proceedings 2016 IEEE Congress on Evolutionary Computation (CEC)* 1227–1232 (2016).
71. Shen, R., Zhan, Q., Wang, Y. & Ma, H. Depression detection by analysing eye movements on emotional images. In *Proceedings ICASSP 2021–2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 7973–7977 (2021).
72. Pan, Z., Ma, H., Zhang, L. & Wang, Y. Depression detection based on reaction time and eye movement. In *Proceedings 2019 IEEE International Conference on Image Processing (ICIP)* 2184–2188 (2019).
73. Le, C., Ma, H. & Wang, Y. A method for extracting eye movement and response characteristics to distinguish depressed people. In *Proceedings Image and Graphics: 9th International Conference, ICIG 2017, Shanghai, China, September 13–15, 2017 Revised Selected Papers, Part I* 9. 489–500 (2017).
74. Zhang, D. et al. Effective differentiation between depressed patients and controls using discriminative eye movement features. *J. Affect. Disord.* **307**, 237–243 (2022).
75. Diao, Y. et al. A combination of P300 and eye movement data improves the accuracy of auxiliary diagnoses of depression. *J. Affect. Disord.* **297**, 386–395 (2022).
76. Shao, W. et al. A multi-modal gait analysis-based detection system of the risk of depression. *IEEE J. Biomed. Health Inform.* **26**, 4859–4868 (2021).
77. Lu, H., Shao, W., Ngai, E., Hu, X. & Hu, B. A new skeletal representation based on gait for depression detection. In *Proceedings 2020 IEEE International Conference on E-health Networking, Application & Services (HEALTHCOM)* 1–6 (2020).
78. Wang, T. et al. A gait assessment framework for depression detection using kinect sensors. *IEEE Sens. J.* **21**, 3260–3270 (2020).

79. Li, W., Wang, Q., Liu, X. & Yu, Y. Simple action for depression detection: using kinect-recorded human kinematic skeletal data. *BMC Psychiatry* **21**, 205 (2021).
80. Yu, Y. et al. Depression and severity detection based on body kinematic features: using kinect recorded skeleton data of simple action. *Front. Neurol.* **13**, 905917 (2022).
81. Wang, Y., Wang, J., Liu, X. & Zhu, T. Detecting depression through gait data: examining the contribution of gait features in recognizing depression. *Front. Psychiatry* **12**, 661213 (2021).
82. Zhang, X. et al. Multimodal depression detection: fusion of electroencephalography and paralinguistic behaviors using a novel strategy for classifier ensemble. *IEEE J. Biomed. Health Inform.* **23**, 2265–2275 (2019).
83. Chen, T., Hong, R., Guo, Y., Hao, S. & Hu, B. MS²-GNN: exploring GNN-based multimodal fusion network for depression detection. *IEEE Trans. Cybern.* **53**, 7749–7759 (2022).
84. Qayyum, A., Razzak, I., Tanveer, M., Mazher, M. & Alhaqbani, B. High-density electroencephalography and speech signal based deep framework for clinical depression diagnosis. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **20**, 2587–2597 (2023).
85. Ahmed, S., Yousuf, M. A., Monowar, M. M., Hamid, M. A. & Alassafi, M. Taking all the factors we need: a multimodal depression classification with uncertainty approximation. *IEEE Access* **11**, 99847–99861 (2023).
86. Zhu, J. et al. Content-based multiple evidence fusion on EEG and eye movements for mild depression recognition. *Comput. Methods Prog. Biomed.* **226**, 107100 (2022).
87. Othmani, A., Zeghina, A.-O. & Muzammel, M. A model of normality inspired deep learning framework for depression relapse prediction using audiovisual data. *Comput. Methods Prog. Biomed.* **226**, 107132 (2022).
88. Yang, L., Jiang, D. & Sahli, H. Integrating deep and shallow models for multi-modal depression analysis—hybrid architectures. *IEEE Trans. Affect. Comput.* **12**, 239–253 (2018).
89. Yang, S., Cui, L., Wang, L., Wang, T. & You, J. Enhancing multimodal depression diagnosis through representation learning and knowledge transfer. *Heliyon* **10**, e25959 (2024).
90. Muzammel, M., Salam, H. & Othmani, A. End-to-end multimodal clinical depression recognition using deep neural networks: a comparative analysis. *Comput. Methods Prog. Biomed.* **211**, 106433 (2021).
91. Joshi, J. et al. Multimodal assistive technologies for depression diagnosis and monitoring. *J. Multimodal Use. Interfaces* **7**, 217–228 (2013).
92. Chen, J. et al. IIFDD: Intra and inter-modal fusion for depression detection with multi-modal information from Internet of Medical Things. *Inf. Fusion* **102**, 102017 (2024).
93. Chen, X. et al. MGSN: Depression EEG lightweight detection based on multiscale DGCN and SNN for multichannel topology. *Biomed. Signal Process. Control* **92**, 106051 (2024).
94. Ying, M. et al. EDT: An EEG-based attention model for feature learning and depression recognition. *Biomed. Signal Process. Control* **93**, 106182 (2024).
95. Sun, C., Jiang, M., Gao, L., Xin, Y. & Dong, Y. A novel study for depression detecting using audio signals based on graph neural network. *Biomed. Signal Process. Control* **88**, 105675 (2024).
96. Shao, X., Ying, M., Zhu, J., Li, X. & Hu, B. Achieving EEG-based depression recognition using Decentralized-Centralized structure. *Biomed. Signal Process. Control* **95**, 106402 (2024).
97. Xu, X., Wang, Y., Wei, X., Wang, F. & Zhang, X. Attention-based acoustic feature fusion network for depression detection. *Neurocomputing* **601**, 128209 (2024).
98. Zou, B. et al. Semi-structural automatic preliminary screening of depressive disorders. *IEEE Trans. Affect. Comput.* **14**, 2823–2838 (2022).
99. Chen, D. et al. Comparative efficacy of multimodal AI methods in screening for major depressive disorder: machine learning model development predictive pilot study. *JMIR Form. Res.* **9**, e56057 (2025).
100. Zhou, L., Hu, B. & Guan, Z.-H. MDRA: a multimodal depression risk assessment model using audio and text. *IEEE Signal Process. Lett.* **32**, 2045–2049 (2025).
101. Xie, W. et al. Interpreting depression from question-wise long-term video recording of SDS evaluation. *IEEE J. Biomed. Health Inform.* **26**, 865–875 (2021).
102. Al Hanai, T., Ghassemi, M. M. & Glass, J. R. Detecting depression with audio/text sequence modeling of interviews. In *Proceedings Interspeech 1716–1720* (ISCA, 2018).
103. Toto, E., Tlachac, M., Stevens, F. L. & Rundensteiner, E. A. Audio-based depression screening using sliding window sub-clip pooling. In *Proceedings 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA)* 791–796 (2020).
104. Sardari, S., Nakisa, B., Rastgoo, M. N. & Eklund, P. Audio based depression detection using convolutional autoencoder. *Expert Syst. Appl.* **189**, 116076 (2022).
105. Ding, H. et al. IntervoxNet: a novel dual-modal audio-text fusion network for automatic and efficient depression detection from interviews. *Front. Phys.* **12**, 1430035 (2024).
106. Shen, Y., Yang, H. & Lin, L. Automatic depression detection: an emotional audio-textual corpus and a gru/bilstm-based model. In *Proceedings ICASSP 2022–2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 6247–6251 (2022).
107. Siuly, S. et al. Electroencephalogram (EEG) and its background. In *EEG Signal Analysis and Classification: Techniques and Applications* 3–21 (Springer International Publishing, 2016).
108. Fox, N. A. If it's not left, it's right: electroencephalograph asymmetry and the development of emotion. *Am. Psychol.* **46**, 863 (1991).
109. Thibodeau, R., Jorgensen, R. S. & Kim, S. Depression, anxiety, and resting frontal EEG asymmetry: a meta-analytic review. *J. Abnorm. Psychol.* **115**, 715 (2006).
110. Chen, T. & Guestrin, C. XGBoost: a scalable tree boosting system. In *Proceedings 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 785–794 (ACM, 2016).
111. Ke, G. et al. Lightgbm: a highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* **30** (2017).
112. Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V. & Gulin, A. CatBoost: unbiased boosting with categorical features. *Adv. Neural Inf. Process. Syst.* **31** (2018).
113. Gönen, M. & Alpaydin, E. Multiple kernel learning algorithms. *J. Mach. Learn. Res.* **12**, 2211–2268 (2011).
114. Bucak, S. S., Jin, R. & Jain, A. K. Multiple kernel learning for visual object recognition: a review. *IEEE Trans. Pattern Anal. Mach. Intell.* **36**, 1354–1369 (2013).
115. Lahat, D., Adali, T. & Jutten, C. Multimodal data fusion: an overview of methods, challenges, and prospects. *Proc. IEEE* **103**, 1449–1477 (2015).
116. Litjens, G. et al. A survey on deep learning in medical image analysis. *Med. Image Anal.* **42**, 60–88 (2017).
117. Vaswani, A. et al. Attention is all you need. *Advances in neural information processing systems* **30** (2017).
118. Lin, C.-C., Lin, K., Wang, L., Liu, Z. & Li, L. Cross-modal representation learning for zero-shot action recognition. In *Proceedings IEEE/CVF Conference on Computer Vision and Pattern Recognition* 19978–19988 (IEEE, 2022).
119. Lundberg, S. M. & Lee, S.-I. A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst.* **30** (2017).

120. Fu, T.-c. A review on time series data mining. *Eng. Appl. Artif. Intell.* **24**, 164–181 (2011).
121. Cerutti, S. In the spotlight: biomedical signal processing. *IEEE Rev. Biomed. Eng.* **1**, 8–11 (2008).
122. Cavanagh, J. F., Napolitano, A., Wu, C. & Mueen, A. The patient repository for EEG data+ computational tools (PRED+ CT). *Front. Neuroinform.* **11**, 67 (2017).
123. Schuller, B. et al. AVEC 2011—the first international audio/visual emotion challenge. In *Proceedings Affective Computing and Intelligent Interaction: Fourth International Conference, ACII 2011, Memphis, TN, USA, October 9–12, 2011, Part II* 415–424 (2011).
124. Schuller, B., Valster, M., Eyben, F., Cowie, R. & Pantic, M. AVEC 2012: the continuous audio/visual emotion challenge. In *Proceedings 14th ACM International Conference on Multimodal Interaction* 449–456 (ACM, 2012).
125. Valstar, M. et al. AVEC 2013: the continuous audio/visual emotion and depression recognition challenge. In *Proceedings 3rd ACM International Workshop on Audio/Visual Emotion Challenge* 3–10 (ACM, 2013).
126. Valstar, M. et al. AVEC 2014: the 4th International audio/visual emotion challenge and workshop. In *Proceedings 4th International Workshop on Audio/Visual Emotion Challenge* 3–10 (ACM, 2014).
127. Valstar, M. et al. AVEC 2016: depression, mood, and emotion recognition workshop and challenge. In *Proceedings 6th International Workshop on Audio/Visual Emotion Challenge* 3–10 (ACM, 2016).
128. Ringeval, F. et al. AVEC 2017: real-life depression, and affect recognition workshop and challenge. In *Proceedings 7th Annual Workshop on Audio/Visual Emotion Challenge* 3–9 (ACM, 2017).
129. Ringeval, F. et al. AVEC 2018 workshop and challenge: Bipolar disorder and cross-cultural affect recognition. In *Proceedings 2018 on Audio/Visual Emotion Challenge and Workshop* 3–13 (ACM, 2018).
130. Ringeval, F. et al. AVEC 2019 workshop and challenge: state-of-mind, detecting depression with AI, and cross-cultural affect recognition. In *Proceedings 9th International on Audio/visual Emotion Challenge and Workshop* 3–12 (ACM, 2019).
131. Cai, H. et al. A multi-modal open dataset for mental-disorder analysis. *Sci. Data* **9**, 178 (2022).
132. Stahlschmidt, S. R., Ulfenborg, B. & Synnergren, J. Multimodal deep learning for biomedical data fusion: a review. *Brief. Bioinform.* **23**, bbab569 (2022).
133. Baltrušaitis, T., Ahuja, C. & Morency, L.-P. Multimodal machine learning: a survey and taxonomy. *IEEE Trans. Pattern Anal. Mach. Intell.* **41**, 423–443 (2018).
134. Custers, B. Click here to consent forever: expiry dates for informed consent. *Big Data Soc.* **3**, 2053951715624935 (2016).
135. Flores, R., Tlachac, M., Toto, E. & Rundensteiner, E. Audiface: multimodal deep learning for depression screening. In *Proceedings Machine Learning for Healthcare Conference* 609–630 (2022).
136. Page, M. J. et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* **372**, n71 (2021).
137. Whiting, P. F. et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. *Ann. Intern. Med.* **155**, 529–536 (2011).
138. Jackson, D. & Turner, R. Power analysis for random-effects meta-analysis. *Res. Synth. Methods* **8**, 290–302 (2017).

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Author contributions

All authors contributed to the study conception and design. L.W., C.W., and C.L. developed the protocol with guidance from and under the supervision of J.J.; L.W., C.W., and C.L. searched the electronic databases and conducted backward and forward reference list checking. The study selection process, data extraction, and risk of bias assessment were carried out by Z.S. and Y.B. Data synthesis was conducted by S.Z. and Q.Z. The introduction section was written by C.W.; the methods, results, and discussion sections were written by L.W., T.M., C.L., Y.H., and J.J. revised the manuscript. The article was revised critically for important intellectual content by all authors. All authors have read and approved the manuscript for publication and agree to be accountable for all aspects of the work.

Competing interests

The authors declare no competing interests.

Additional information

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